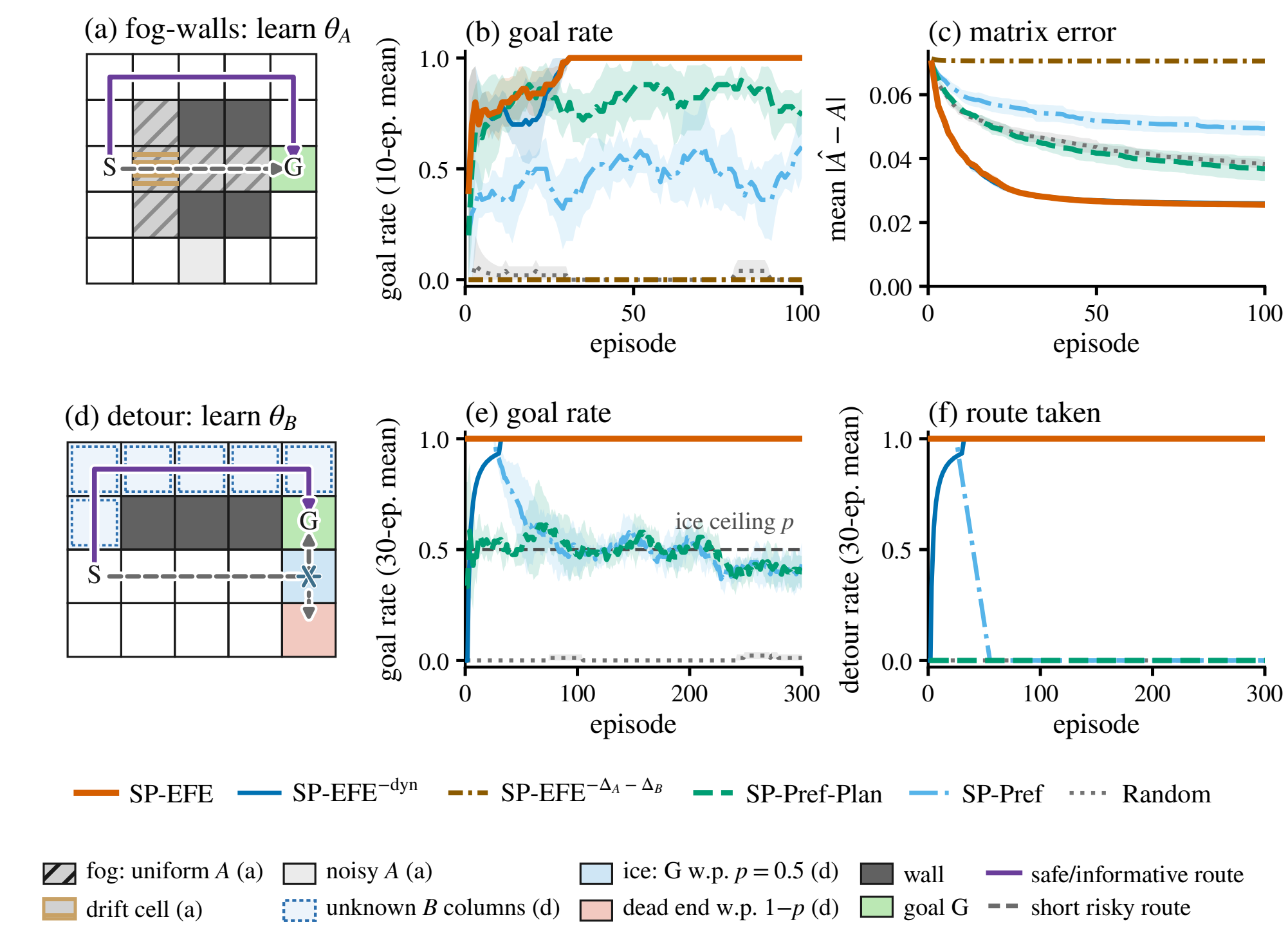

(a) fog-walls: learn θ_A
S
G
(b) goal rate
goal rate (10-ep. mean)
1.0
0.5
0.0
0
50
100
episode
(c) matrix error
mean $|\hat{A} - A|$
0.06
0.04
0.02
0.00
0
50
100
episode
(d) detour: learn θ_B
G
S
(e) goal rate
goal rate (30-ep. mean)
ice ceiling p
1.0
0.5
0.0
0
100
200
300
episode
(f) route taken
detour rate (30-ep. mean)
1.0
0.5
0.0
0
100
200
300
episode
SP-EFE
SP-EFE$^{-\text{dyn}}$
SP-EFE$^{-\Delta_A - \Delta_B}$
SP-Pref-Plan
SP-Pref
Random
fog: uniform A (a)
noisy A (a)
ice: G w.p. $p = 0.5$ (d)
wall
safe/informative route
drift cell (a)
unknown B columns (d)
dead end w.p. $1-p$ (d)
goal G
short risky route